

Week_3_Mini project

September 12, 2026

1 Day 7 — Mini Project: EDA & Visualization Report

1.1 Objective

The objective of this mini project is to perform Exploratory Data Analysis (EDA) and create meaningful visualizations using the Customer Churn dataset.

The analysis includes summary statistics, outlier detection, data distribution, categorical comparison, relationships between numerical variables, and correlation analysis.

The main goal is to understand the dataset and communicate important findings using appropriate charts.

```
[2]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_excel(
    "Day_4_Ensemble_Learning_Customer_Churn.xlsx",
    sheet_name="Customer Churn Data"
)

df
```

```
[2]:
```

	CustomerID	Age	Tenure_Months	MonthlyCharges	Contract	\
0	1001	25	3	65	Month-to-month	
1	1002	34	18	72	One year	
2	1003	45	36	85	Two year	
3	1004	29	7	55	Month-to-month	
4	1005	52	48	95	Two year	
5	1006	41	24	78	One year	
6	1007	23	2	50	Month-to-month	
7	1008	38	30	88	One year	
8	1009	49	42	92	Two year	
9	1010	31	12	60	Month-to-month	
10	1011	27	8	68	Month-to-month	
11	1012	56	60	105	Two year	
12	1013	44	28	82	One year	
13	1014	36	20	75	One year	

14	1015	30	15	58	Month-to-month
15	1016	47	40	90	Two year
16	1017	22	4	45	Month-to-month
17	1018	39	32	80	One year
18	1019	53	54	100	Two year
19	1020	33	16	70	Month-to-month

	InternetService	TechSupport	PaymentMethod	Churn
0	Fiber optic	No	Electronic check	Yes
1	DSL	Yes	Bank transfer	No
2	Fiber optic	Yes	Credit card	No
3	DSL	No	Electronic check	Yes
4	Fiber optic	Yes	Credit card	No
5	DSL	Yes	Bank transfer	No
6	No	No	Mailed check	Yes
7	Fiber optic	Yes	Electronic check	No
8	Fiber optic	Yes	Credit card	No
9	DSL	No	Mailed check	Yes
10	Fiber optic	No	Electronic check	Yes
11	No	Yes	Bank transfer	No
12	DSL	Yes	Credit card	No
13	Fiber optic	No	Electronic check	Yes
14	DSL	No	Mailed check	Yes
15	Fiber optic	Yes	Credit card	No
16	No	No	Mailed check	Yes
17	DSL	Yes	Bank transfer	No
18	Fiber optic	Yes	Credit card	No
19	DSL	No	Electronic check	Yes

1.2 1. Summary Statistics

Summary statistics are used to understand the numerical variables in the dataset.

The `describe()` function provides important statistical measures such as count, mean, standard deviation, minimum, quartiles, and maximum.

This helps us understand the central tendency and spread of the numerical data.

```
[3]: df.describe()
```

```
[3]:
```

	CustomerID	Age	Tenure_Months	MonthlyCharges
count	20.00000	20.000000	20.000000	20.000000
mean	1010.50000	37.700000	24.950000	75.650000
std	5.91608	10.387746	17.479236	16.865646
min	1001.00000	22.000000	2.000000	45.000000
25%	1005.75000	29.750000	11.000000	63.750000
50%	1010.50000	37.000000	22.000000	76.500000
75%	1015.25000	45.500000	37.000000	88.500000

```
max      1020.00000  56.000000      60.000000      105.000000
```

1.3 2. Outlier Detection using IQR

An outlier is a value that is unusually high or low compared with the rest of the data.

The Interquartile Range (IQR) method is used to identify potential outliers.

$IQR = Q3 - Q1$

$Lower\ Bound = Q1 - 1.5 \times IQR$

$Upper\ Bound = Q3 + 1.5 \times IQR$

In this project, MonthlyCharges is selected for outlier analysis.

```
[4]: Q1 = df["MonthlyCharges"].quantile(0.25)
      Q3 = df["MonthlyCharges"].quantile(0.75)

      IQR = Q3 - Q1

      lower_bound = Q1 - 1.5 * IQR
      upper_bound = Q3 + 1.5 * IQR

      print("Q1:", Q1)
      print("Q3:", Q3)
      print("IQR:", IQR)
      print("Lower Bound:", lower_bound)
      print("Upper Bound:", upper_bound)
```

```
Q1: 63.75
```

```
Q3: 88.5
```

```
IQR: 24.75
```

```
Lower Bound: 26.625
```

```
Upper Bound: 125.625
```

1.3.1 Identifying Outliers

Values below the lower bound or above the upper bound are considered potential outliers.

```
[5]: outliers = df[
      (df["MonthlyCharges"] < lower_bound) |
      (df["MonthlyCharges"] > upper_bound)
    ]

      print("Number of Outliers:", len(outliers))

      outliers
```

```
Number of Outliers: 0
```

```
[5]: Empty DataFrame
     Columns: [CustomerID, Age, Tenure_Months, MonthlyCharges, Contract,
     InternetService, TechSupport, PaymentMethod, Churn]
     Index: []
```

1.4 3. Box Plot — Outlier Visualization

A Box Plot is used to visualize the distribution of numerical data and identify potential outliers.

In this project, a box plot is created for MonthlyCharges.

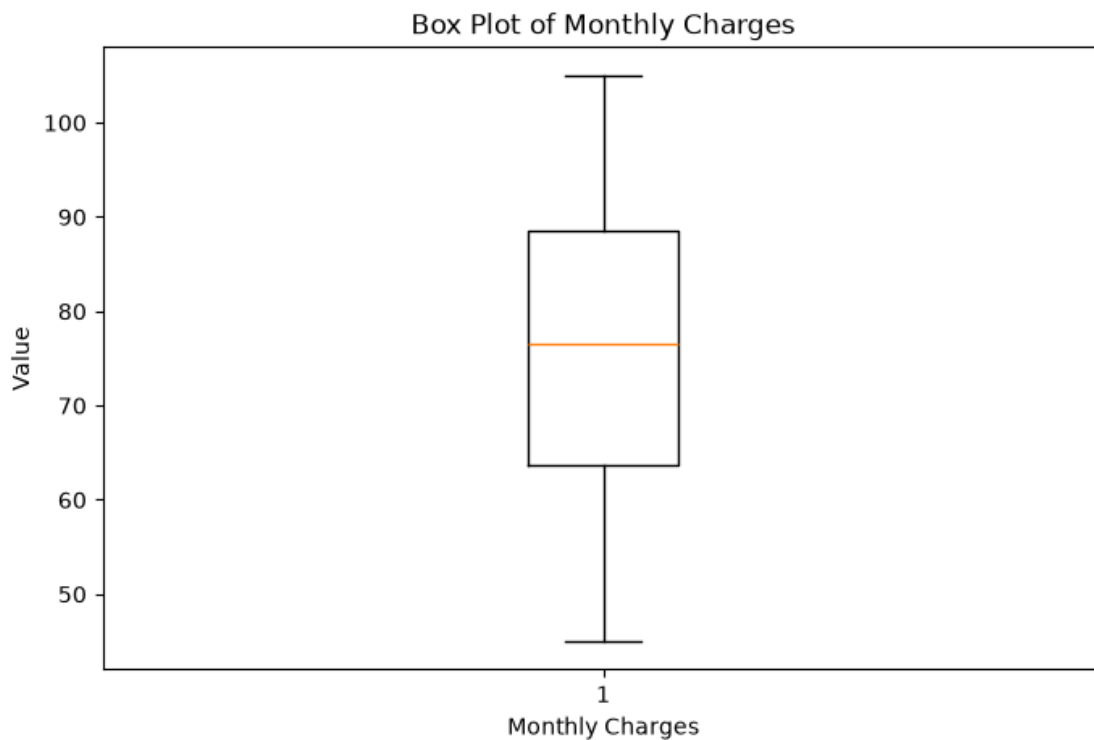
The box plot provides a visual representation of the median, quartiles, spread, and potential outliers.

```
[6]: plt.figure(figsize=(8, 5))

plt.boxplot(df["MonthlyCharges"])

plt.title("Box Plot of Monthly Charges")
plt.xlabel("Monthly Charges")
plt.ylabel("Value")

plt.show()
```



1.5 4. Histogram — Age Distribution

A Histogram is used to understand the distribution of numerical data.

In this project, the distribution of customer Age is visualized.

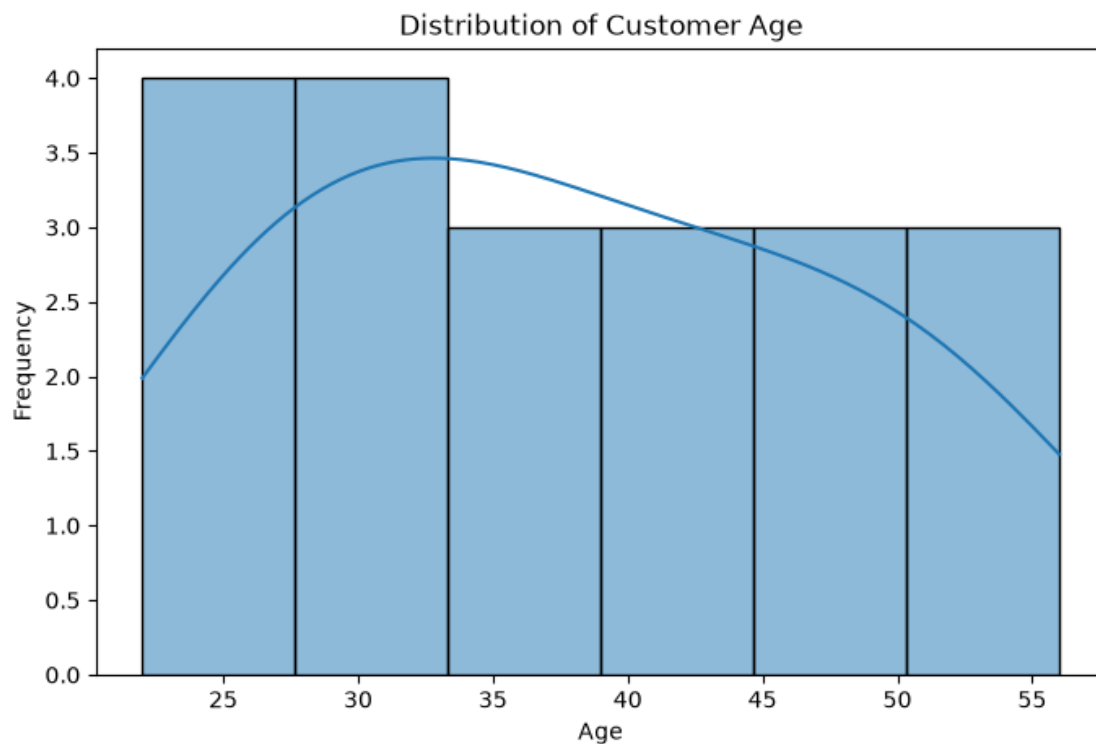
The histogram helps us understand how customer ages are spread across different ranges.

```
[7]: plt.figure(figsize=(8, 5))

sns.histplot(
    data=df,
    x="Age",
    bins=6,
    kde=True
)

plt.title("Distribution of Customer Age")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```



1.6 5. Bar Chart — Contract Type Comparison

A Bar Chart is used to compare values across different categories.

In this project, the number of customers in each Contract type is visualized.

This helps us compare the distribution of customers across different contract categories.

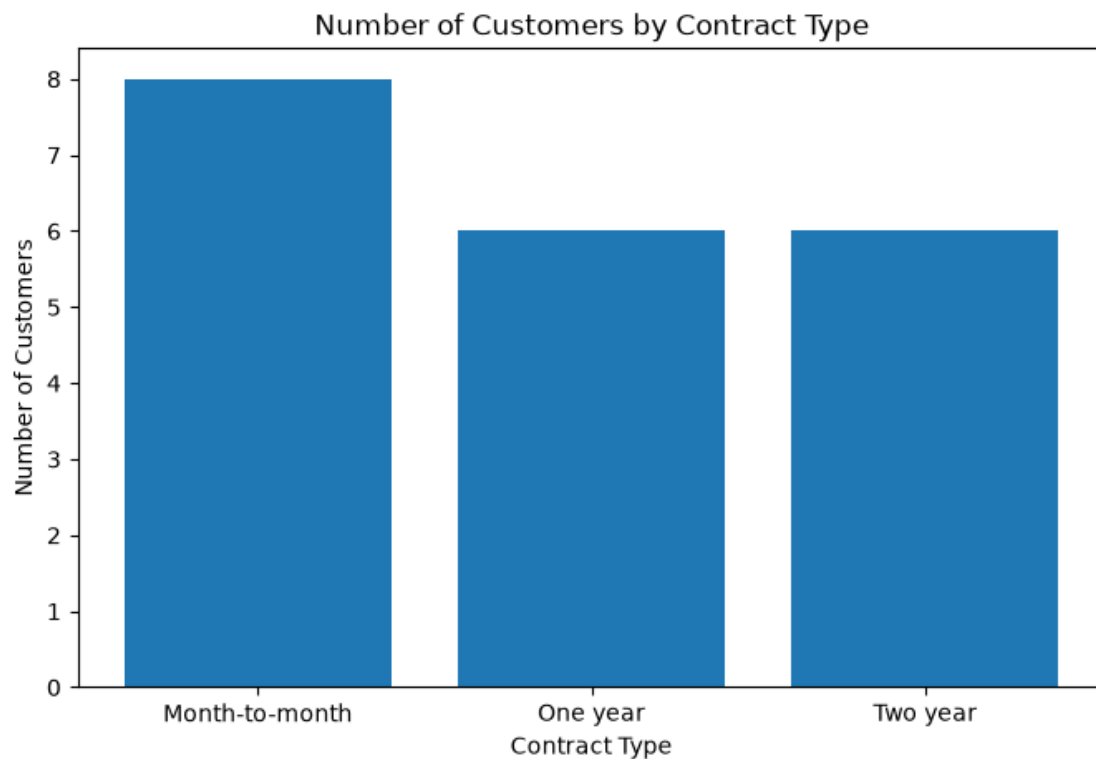
```
[8]: contract_counts = df["Contract"].value_counts()

plt.figure(figsize=(8, 5))

plt.bar(
    contract_counts.index,
    contract_counts.values
)

plt.title("Number of Customers by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")

plt.show()
```



1.7 6. Scatter Plot — Age vs Monthly Charges

A Scatter Plot is used to understand the relationship between two numerical variables.

In this project, Age and MonthlyCharges are compared using a scatter plot.

Each point represents one customer.

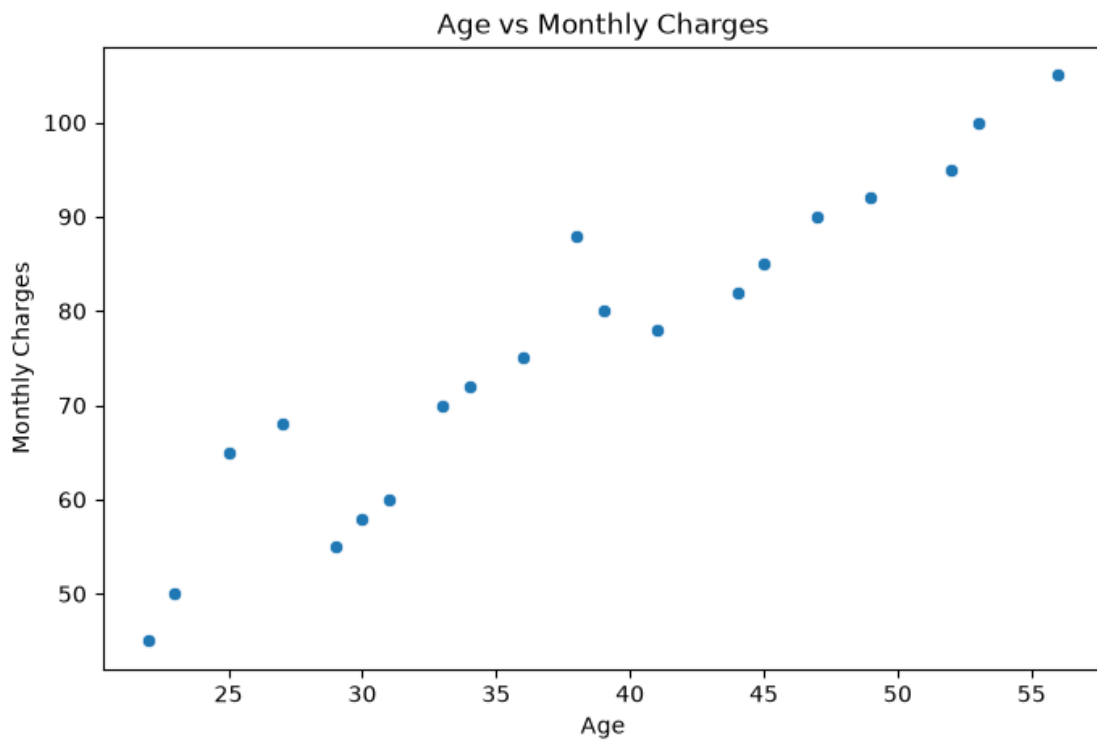
This visualization helps us observe whether there is any visible relationship between customer age and monthly charges.

```
[9]: plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="Age",
    y="MonthlyCharges"
)

plt.title("Age vs Monthly Charges")
plt.xlabel("Age")
plt.ylabel("Monthly Charges")

plt.show()
```



1.8 7. Correlation Heatmap

A Correlation Heatmap is used to identify relationships between numerical variables.

Correlation values range from -1 to +1.

A value close to +1 indicates a strong positive relationship, while a value close to -1 indicates a strong negative relationship.

A value close to 0 indicates a weak or no linear relationship.

The heatmap makes these relationships easier to understand visually.

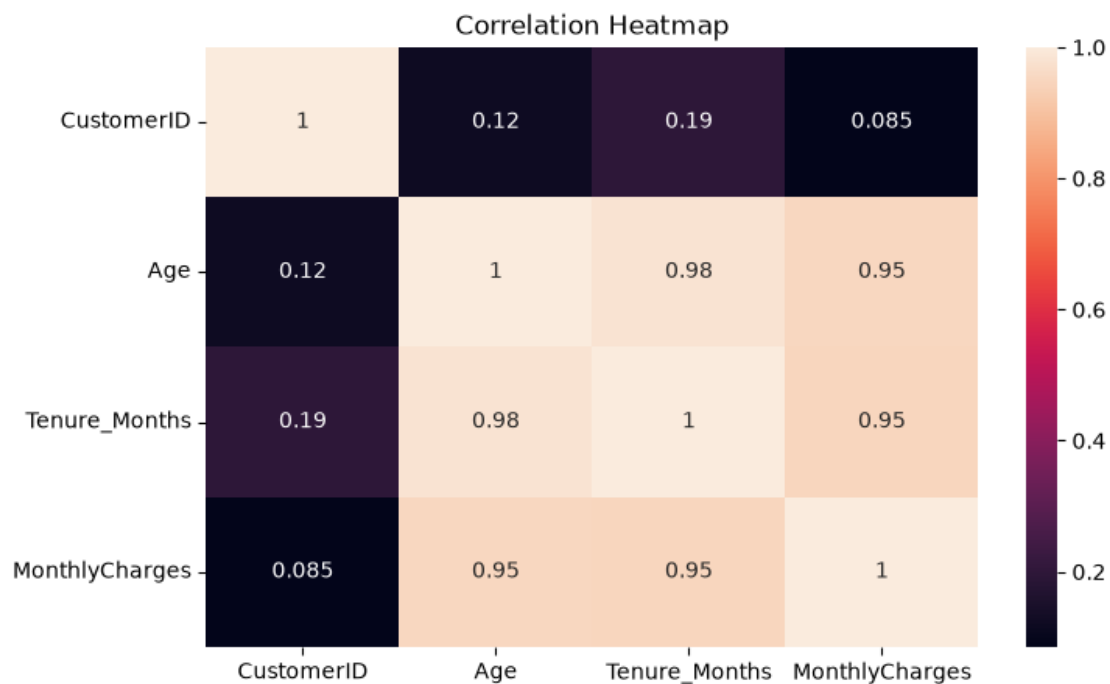
```
[10]: correlation = df.corr(numeric_only=True)

plt.figure(figsize=(8, 5))

sns.heatmap(
    correlation,
    annot=True
)

plt.title("Correlation Heatmap")

plt.show()
```



1.9 8. Key Insights

Based on the EDA and visualizations, the following insights were observed:

- The dataset contains 20 customer records and 9 columns.
- There are no missing values in the dataset.
- The average customer age is approximately 37.7 years.
- The average MonthlyCharges is approximately 75.65.
- The IQR analysis of MonthlyCharges does not identify any potential outliers.

1.10 Conclusion

In this mini project, Exploratory Data Analysis and Data Visualization were performed on the Customer Churn dataset.

Summary statistics were generated to understand the numerical variables. The IQR method and Box Plot were used for outlier analysis.

Histogram, Bar Chart, Scatter Plot, and Correlation Heatmap were used to understand data distribution, category comparison, numerical relationships, and correlation between variables.

The analysis helped in understanding the structure, distribution, relationships, and important characteristics of the customer data.

Overall, EDA and visualization provide useful insights from raw data and help in making better data-driven decisions.